

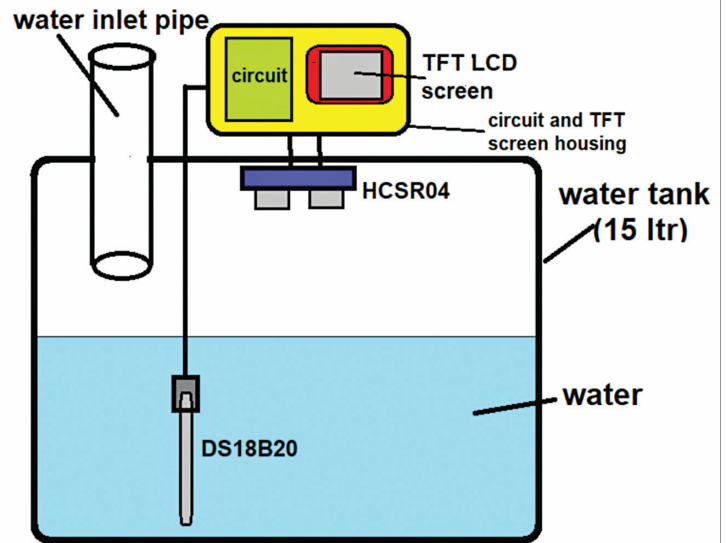
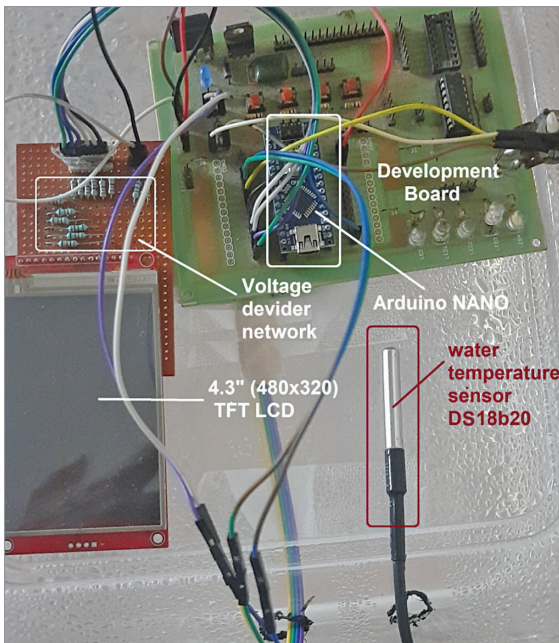


Fig. 5: Assembly diagram

TABLE 2
TFT LCD TO ARDUINO
NANO PIN MAPPING

TFT LCD Pin	Arduino Nano Pin
Vcc	5V
GND	GND
CS	D8
RST	D9
DC	D10
SDO	D11
SDI	D12
SCK	D13

TFT LCD input pins, as shown in the circuit diagram. This is necessary because the TFT LCD operates at 3.3V logic levels, whereas the Arduino Nano outputs 5V logic signals. The resistor network reduces the 5V signal from the Arduino to a safe 3.3V level for the TFT. However, some TFT displays support both 3.3V and 5V logic levels. In such cases, the resistor network may not be required; check the display's datasheet for confirmation.

The HC-SR04 ultrasonic sensor has four interfacing pins: (1) Vcc, (2) Trig, (3) Echo, and (4) GND. The Vcc pin is connected to the 5V output of the Arduino board, and

the GND pin is connected to the circuit ground. The Trig and Echo pins are connected to digital pins D4 and D3 of the Arduino, respectively.

Software

The Arduino IDE platform is used to develop the code for this system. Several libraries are required, including DallasTemperature.h to read temperature data from the DS18B20 sensor, OneWire.h to initialise the one-wire protocol for communication with the DS18B20, LCDWIKI_GUI.h to display text, numbers, shapes, animations, and GUI elements on the TFT screen, and LCDWIKI_SPI.h to facilitate communication with the SPI-based TFT display using the 4-wire SPI protocol.

These libraries must be installed in the Arduino IDE. To install them, open the Arduino IDE, go to Tools > Manage Libraries, search for the required libraries, and install them. For the LCDWIKI libraries, download the ZIP file from https://github.com/gitcnd/LCDWIKI_SPI. After downloading, add the library manually via Sketch > Include Library > Add .ZIP Library.

In the code, the Arduino uses the

LCDWIKI_GUI and LCDWIKI_SPI libraries to display water level and temperature on a 480 × 320 pixel SPI TFT LCD screen. These libraries allow the Arduino to display text, create graphical elements such as a water tank and temperature bar graph, and generate animations—for example, water filling the tank or temperature rising in the bar graph.

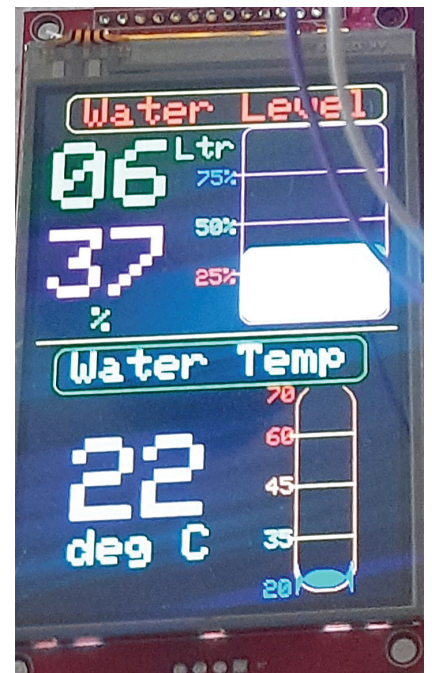


Fig. 6: Display of the water level and temperature